
The Biology of Agentic Research Engineering: An Agent-Driven Variant Effect Prediction Workshop

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Abstract

Tool-calling is necessary but insufficient for biological research engineering. Agents can shell out to bioinformatics CLIs, but the surface they land on is hostile: stateful pipelines, undocumented config conventions, and silent failure modes that look like success. We argue that the missing piece is a *harness*: opinionated geometric primitives, reproducibility scaffolding, and agent-friendly APIs over normally-unfriendly biological surfaces. We demonstrate this with a worked variant effect prediction (VEP) example driven end-to-end from a natural-language prompt. On 500 ClinVar missense variants across 400 disease genes, an ESM-1b zero-shot log-likelihood ratio reaches AUROC 0.930 (95% CI [0.906, 0.951]) — within statistical noise of the 0.905 literature anchor from Brandes et al. [2023] on $n=36,537$. The harness, the experiment, and this paper were produced by an agent operating against the repository the reader can clone today.

1 Introduction

Classifying missense variants as pathogenic or benign is a prerequisite for clinical genetics, yet most variants in ClinVar [Landrum et al., 2018] remain of uncertain significance. Protein language models such as ESM-1b [Rives et al., 2021] assign zero-shot likelihoods to amino-acid substitutions, providing a strong unsupervised baseline that does not require labeled training data [Brandes et al., 2023]. The barrier is not the model but the surrounding pipeline: data loaders, tokenizers, truncation policies, evaluation harnesses, and figure code typically demand bespoke glue. We demonstrate that a well-designed harness collapses that glue into reusable primitives the agent can drive from a single prompt.

2 Related Work

Protein language models. ESM-1 [Rives et al., 2021] and its variants established that masked language-model pretraining at protein scale yields embeddings with biological structure; ESM-2 extended this to larger model and data scales [Lin et al., 2023]. Brandes et al. [2023] showed that ESM-1b’s zero-shot likelihood ratios at the variant position reproduce clinical labels on ClinVar at AUROC 0.905, establishing the VEP scoring rule we adopt.

Cross-modal foundation models. Evo [Nguyen et al., 2024] and related DNA-scale models extend foundation- model training to the nucleotide modality, motivating multi-modal harness designs.

3 Methods

We score each missense substitution with the masked log-likelihood ratio (LLR) defined by Brandes et al. [2023]. The wild-type protein sequence is passed through ESM-1b once, and the LLR is read

off the softmax at the variant position as

$$\text{LLR} = \log P(\text{mut} \mid \text{WT}) - \log P(\text{wt} \mid \text{WT}). \quad (1)$$

Both probabilities are conditioned on the wild-type context, in a single forward pass — there is no mutant-sequence forward. Negative LLRs indicate that the model finds the mutant residue less likely than the wild-type at that position. We follow Brandes’s convention *negative = deleterious*; the AUROC predictor is $-\text{LLR}$ so that pathogenic is the positive class.

For proteins longer than ESM-1b’s 1022-residue context, we apply Brandes et al.’s variant-centered truncation (their option 4): a single $w=1022$ window centered on the variant position. Their Extended Data Fig. 6a shows this is within noise of their sliding-window weighted-average aggregation at the same window size.

The workshop validation set is 500 ClinVar missense variants (250 pathogenic, 250 benign) across 400 disease genes, drawn under the v2 spec recorded in `experiments/notebooks/data/workshop_set_manifest.json` — matching Brandes et al.’s label-scope binarization (ClinSigSimple) so the AUROC is directly comparable.

4 Results

The LLR scorer reaches AUROC 0.930 on the workshop set (95% CI [0.906, 0.951] via 10k-resample bootstrap, seed 42); Brandes et al.’s 0.905 on $n=36,537$ sits within our CI. Figure 1 renders the resolution ladder: panel A is the single demo-pair LLR (the prototype beat), panel B is the ROC over the $n=500$ workshop set (the validate beat), and panel C is the literature anchor at $n=36,537$ (the ceiling).

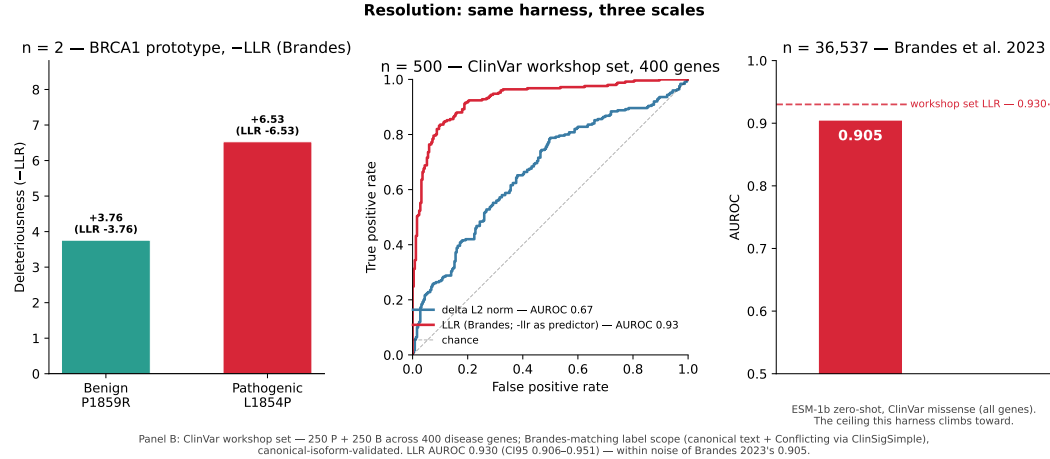


Figure 1: Resolution ladder. (A) Single-variant demo: BRCA1 L1854P pathogenic vs P1859R benign LLRs via ESM-1b; the prototype beat. (B) ROC curves over the $n=500$ workshop set; LLR AUROC 0.930 vs delta-norm AUROC 0.672. (C) Brandes et al. [2023]’s ESM-1b zero-shot AUROC 0.905 on $n=36,537$ ClinVar missense; the workshop floor (LLR at the displayed layer) is pulled forward as a dashed reference line.

5 Discussion

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